

DIS Department of Mathematics
MYP Grade 10 Extended – Basics of Trigonometry Assessment

Name: Answers Date: _____

You will be graded against criteria A (no calculator) and D (GDC required). Unless stated otherwise all final answers are to be given exactly or correct to three significant figures.

Formulas:

Reciprocal trigonometric ratios:

$$\csc\theta = \frac{1}{\sin\theta} \quad \sec\theta = \frac{1}{\cos\theta} \quad \cot\theta = \frac{1}{\tan\theta}$$

Converting from radians to degrees: $\times \frac{180}{\pi}$

Converting from degrees to radians: $\times \frac{\pi}{180}$

Circumference of a circle: $C = 2\pi r$

Area of a circle: $A = \pi r^2$

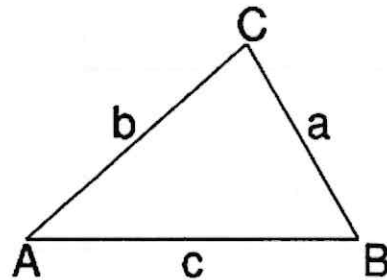
Arc length: $l = \theta r$ (θ in radians) and $l = \frac{\pi r \theta}{180^\circ}$ (θ in degrees)

Sector area: $A = \frac{1}{2} \theta r^2$ (θ in radians) and $A = \frac{\pi r^2 \theta}{360^\circ}$ (θ in degrees)

Area of any triangle: $A = \frac{1}{2} ab \sin C$

Sine Rule: $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

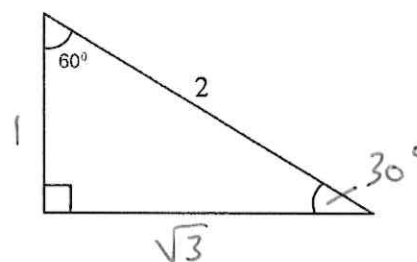
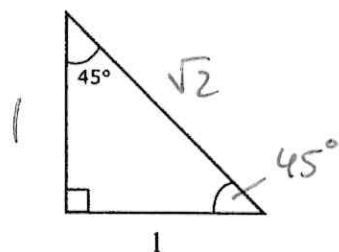
Cosine-Rule: $c^2 = a^2 + b^2 - 2ab \cos C$



Pythagorean Identity: $\sin^2\theta + \cos^2\theta = 1$

Criterion A questions (no calculator)

1. Fill in all missing sides and angles in the triangles below.



2. Complete the table below.

Angle in degrees	Angle in radians
30°	$\frac{\pi}{6}$
60°	$\frac{\pi}{3}$
100°	$\frac{5\pi}{9}$
144°	$\frac{4\pi}{5}$
90°	0.5π
$\left(\frac{180}{\pi}\right)^\circ$	1
270°	$\frac{3\pi}{2}$

3. Evaluate the following expressions. Give all answers in simplified form as integers or fractions with positive integer denominators.

(a) $\cos(180^\circ) = -1$

(b) $\tan\left(\frac{2\pi}{3}\right) = -\sqrt{3}$

(c) $\sin(225^\circ) = -\frac{\sqrt{2}}{2}$

(d) $\tan\left(\frac{7\pi}{4}\right) = -1$

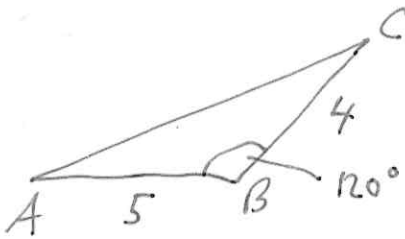
(e) $\csc\left(-\frac{\pi}{2}\right) = -1$

(f) $\cos(240^\circ) = -\frac{1}{2}$

(g) $\cot(300^\circ) = -\frac{\sqrt{3}}{3}$

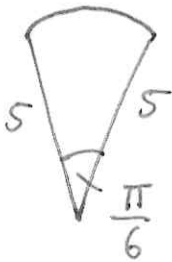
(i) $\frac{\sec\left(\frac{\pi}{6}\right)}{\sin(150^\circ)} = \frac{\frac{2\sqrt{3}}{3}}{\frac{1}{2}} = \frac{2\sqrt{3}}{3} \cdot 2 = \underline{\underline{\frac{4\sqrt{3}}{3}}}$

4. A triangle ABC has side lengths AB = 5 cm, BC = 4 cm, and angle B = 120° . Find the area of this triangle, giving your answer in the form $a\sqrt{b}$, where $a, b \in \mathbb{Z}^+$.



$$\begin{aligned} \text{Area} &= \frac{1}{2} \cdot 5 \cdot 4 \sin 120^\circ \\ &= 10 \cdot \frac{\sqrt{3}}{2} = \underline{\underline{5\sqrt{3} \text{ cm}^2}} \end{aligned}$$

5. Calculate the **perimeter** of a sector with angle $\theta = \frac{\pi}{6}$ and circle diameter 10 cm.



$$\begin{aligned} \text{perimeter} &= 2 \cdot 5 + \frac{\pi}{6} \cdot 5 \\ &= 10 + \frac{5\pi}{6} \text{ cm} \end{aligned}$$

6. A sector has angle $\theta = \frac{\pi}{3}$, and area $\frac{\pi}{3} \text{ cm}^2$. Find the sector's radius.



$$\begin{aligned} \frac{\pi}{3} &= \frac{1}{2} \cdot \frac{\pi}{3} \cdot r^2 \\ r^2 &= 2 \\ \therefore r &= \underline{\underline{\sqrt{2} \text{ cm}}} \end{aligned}$$

7. Given that $\sin \theta = \frac{2}{5}$, find the two possible values of $\cos \theta$. Write your answer in simplest form as a fraction with a positive integer denominator.

$$\begin{aligned} \left(\frac{2}{5}\right)^2 + \cos^2 \theta &= 1 \rightarrow \cos^2 \theta = \frac{21}{25} \\ \therefore \cos \theta &= \underline{\underline{\pm \frac{\sqrt{21}}{5}}} \end{aligned}$$

8. If $\cos\theta = -\frac{1}{3}$ and given that $\tan\theta < 0$, find the value of $\sin\theta$.

$$\left(-\frac{1}{3}\right)^2 + \sin^2\theta = 1 \rightarrow \sin^2\theta = \frac{8}{9}$$

$$\therefore \sin\theta = \pm \frac{2\sqrt{2}}{3}$$

But $\tan\theta < 0 \therefore$ 2nd quadrant
 $\therefore \sin\theta = \underline{\underline{\frac{2\sqrt{2}}{3}}}$

9. Given that $\sec\theta = 2$, find the two possible values of $\tan\theta$.

$$\frac{1}{\cos\theta} = 2 \rightarrow \cos\theta = \frac{1}{2}$$

$$\left(\frac{1}{2}\right)^2 + \sin^2\theta = 1 \rightarrow \sin^2\theta = \frac{3}{4}$$

$$\therefore \sin\theta = \underline{\underline{\pm \frac{\sqrt{3}}{2}}}$$

$$\therefore \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\pm \frac{\sqrt{3}}{2}}{1/2} = \underline{\underline{\pm \sqrt{3}}}$$

Section A

1. (a) **METHOD 1**

attempt to use the cosine rule

(M1)

$$\cos \theta = \frac{4^2 + 4^2 - 5^2}{2 \times 4 \times 4} \text{ (or equivalent)}$$

A1

$$\theta = 1.35$$

A1

[3 marks]

METHOD 2

attempt to split triangle AOB into two congruent right triangles

(M1)

$$\sin\left(\frac{\theta}{2}\right) = \frac{2.5}{4}$$

A1

$$\theta = 1.35$$

A1

[3 marks]

(b) attempt to find the area of the shaded region

(M1)

$$\frac{1}{2} \times 4 \times 4 \times (2\pi - 1.35\dots)$$

A1

$$= 39.5 \text{ (cm}^2\text{)}$$

A1

[3 marks]

Total [6 marks]

2. (a) $\left(1 + \frac{5.5}{4 \times 100}\right)^4$
1.056

(M1)(A1)

A1

[3 marks]

continued...

7

The diagram below shows two circles with centre O and radii 3 cm and 5 cm. The points P and Q lie on the larger circle and angle POQ = x , where $0 < x < \frac{\pi}{2}$.

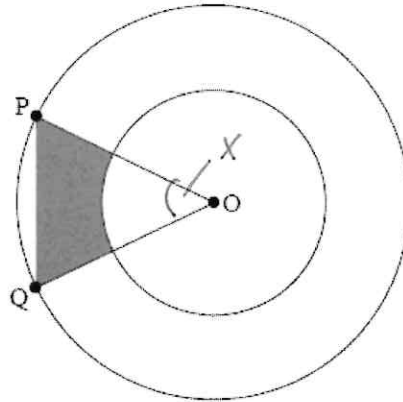


diagram not to scale

Given that the area of the shaded region is 3.8938 cm^2 , find the value of x .

$$\text{Area} = \underbrace{\frac{1}{2} \cdot 5^2 \sin x}_{\text{area of triangle OPQ}} - \underbrace{\frac{1}{2} x \cdot 3^2}_{\substack{\text{area of} \\ \text{sector} \\ \text{with } r=3 \text{ cm}}} = 3.8938$$

$$\therefore \underline{\underline{x = 0.524}}$$

4. attempting to use the area of sector formula (including for a semicircle) **M1**
 semi-circle $\frac{1}{2} \pi \times 5^2 = \frac{25\pi}{2} = 39.26990817\dots$ **(A1)**
 angle in smaller sector is $\pi - \theta$ **(A1)**
 area of sector $= \frac{1}{2} \times 2^2 \times (\pi - \theta)$ **(A1)**
 attempt to total a sum of areas of regions to 44 **(M1)**
 $2(\pi - \theta) = 44 - 39.26990817\dots$
 $\theta = 0.777 \left(= \frac{29\pi}{4} - 22 \right)$ **A1**

Note: Award all marks except the final **A1** for correct working in degrees.

Note: Attempt to solve with goat inside triangle should lead to nonsense answer and so should only receive a maximum of the two **M** marks.

[6 marks]

5. (a) $f(-x) = \frac{3(-x)^2 + 10}{(-x)^2 - 4}$ **A1**
 $= \frac{3x^2 + 10}{x^2 - 4} = f(x)$
 $f(x) = f(-x)$ **R1**
 hence this is an even function **AG**

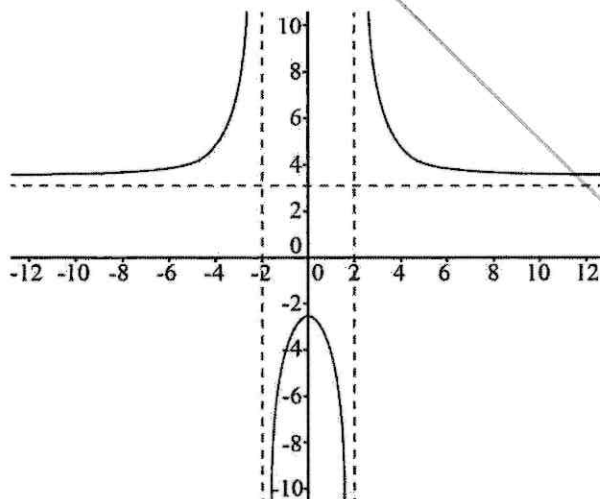
Note: Award **A1R1** for the statement, all the powers are even hence $f(x) = f(-x)$.

Note: Just stating all the powers are even is **A0R0**.

Note: Do not accept arguments based on the symmetry of the graph.

[2 marks]

- (b) (i)



correct shape in 3 parts which are asymptotic and symmetrical **A1**
 correct vertical asymptotes clear at 2 and -2 **A1**
 correct horizontal asymptote clear at 3 **A1**

QUESTION 16

$$\frac{\sin C}{7} = \frac{\sin 30^\circ}{5}$$

$$\Rightarrow \sin C = \frac{7 \sin 30^\circ}{5}$$

M1A1

$$C = 44.4^\circ$$

A1

$$\text{or } C = 135.6^\circ$$

A1

$$\Rightarrow B = 105.6^\circ \text{ or } 14.4^\circ$$

A1

$$\begin{aligned} \Rightarrow \text{Difference in area } \triangle ABC &= \frac{1}{2} ac (\sin B_1 - \sin B_2) \\ &= \frac{1}{2} (5)(7)(\sin 105.6^\circ - \sin 14.4^\circ) \\ &= 12.5 \text{ cm}^2 \end{aligned}$$

A1

N2

Note: There are several ways of solving this problem which require the acute value of C to be found. Award *M1A1A1* for this and then *A1A1A1* for what follows.

QUESTION 17

$$\int \frac{1}{y} dy = \int \frac{4x}{(x+2)^2} dx$$

M1

$$\text{Let } u = x+2 \Rightarrow x = u-2$$

$$du = dx$$

M1

$$\ln y = \int \frac{4(u-2)}{u^2} du$$

$$= \int \frac{4}{u} - \frac{8}{u^2} du$$

A1

$$\ln y = 4 \ln u + 8u^{-1} + c$$

A1

$$\ln y = 4 \ln(x+2) + \frac{8}{x+2} + c$$

A1

$$(-1, 1) \Rightarrow c = -8$$

A1

$$\Rightarrow \ln y = 4 \ln(x+2) + \frac{8}{x+2} - 8$$

N0

